

Django Tutorial Notes: Templates & Inheritance

Topic #16: Master Layout, Static Files, Navigation Bar & CSRF Security (Hinglish Study Notes)

💡 Introduction (Overview)

Is lecture mein humne sikha ki Django projects mein **Template Inheritance** aur **Static Files (CSS, JS, Images)** ko properly kaise manage kiya jata hai. Code duplication avoid karne ke liye Base Template banana aur Navbar include karna bahut important hai.

1. Project Setup & Folder Structure

Project aur app create karne ke baad folders ko properly organize karna padta hai taaki Django files ko easily detect kar paaye:

- **Project Command:** `django-admin startproject myproject8`
- **App Command:** `python3 manage.py startapp blog`
- **Templates Folder (Root):** Isme `base.html`, `home.html`, aur `nav_bar.html` rakhte hain.
- **Static Folder (Root):** Iske andar `css/style.css`, `js/scripts.js`, aur `images/logo.png` rakhe jaate hain.

2. Settings Configuration (`settings.py`)

Django ko batana padta hai ki humari templates aur static files kahan par hain:

- **INSTALLED_APPS:** Apni app ka naam (jaise `'blog'`) yahan register karna zaroori hai.
- **TEMPLATES_DIRS:** Yahan root templates folder configure karte hain: `BASE_DIR / 'templates'`.
- **STATICFILES_DIRS:** Static files ke liye path set karte hain: `STATICFILES_DIRS = [BASE_DIR / 'static']`.

3. URL Routing Setup (`urls.py`)

Routing do levels par hoti hai - Main Project aur App level par:

- **Main `urls.py`:** App ke urls ko include karte hain: `path('blog/', include('blog.urls'))`.
- **App `urls.py` (`blog/urls.py`):** Views ko path ke sath map karte hain aur `name` attribute zaroori dete hain taaki dynamic URL tagging ho sake.

4. Template Inheritance & Base Layout

Common layout ko ek baar `base.html` mein likh kar baaki pages mein inherit kiya jata hai:

- **Dynamic Blocks:** `{% block title %}Default Title{% endblock %}` aur `{% block content %}{% endblock %}` use karke child pages apna content inject karte hain.
- **Extending:** Child page (jaise `home.html` ya `about.html`) ke shuru mein likhte hain: `{% extends "base.html" %}`.
- **Component Include:** Navbar jaisi repeated cheezon ko alag file mein rakh kar include karte hain: `{% include "nav_bar.html" %}`.

5. Static Files & Dynamic URL Tagging

Asset Type	Template Syntax (Hinglish Explanation)
Load Static	Template ke shuru mein likhna padta hai: <code>{% load static %}</code>
CSS Link	<code><link rel="stylesheet" href="{% static 'css/style.css' %}"></code>
Image Source	<code></code>
Dynamic URL	Hardcode URL (jaise <code>/about/</code>) ki jagah tag use karo: <code>About</code>

6. Form Security & JavaScript

- **CSRF Token:** Django forms mein `{% csrf_token %}` tag lagana mandatory hota hai. Yeh cross-site request forgery attacks se website ko secure karta hai.
- **Method POST:** Data send karne ke liye form mein `method="POST"` define karte hain.
- **JS Integration:** Static folder ki JavaScript file ko `<script src="{% static 'js/scripts.js' %}"></script>` ke zariye base template mein load kiya jata hai.

Summary

Is tarah se humne Django mein clean folder structure, template inheritance, navigation routing, aur static asset linking ko successfully complete kiya!